SmartSpend — Financial Health Score (FHS) Literature Summary

Prepared for: Ms. Ellen F. Mangaoang, DIT — Capstone Adviser
Prepared by: Directo, Brix A. · Rubis, Cyrille John M. · Madayag, Djaunathan Albert S.
Lorma Colleges — CCSE — BSIT Capstone 2 — AY 2026–2027

What is the Financial Health Score?

The SmartSpend **Financial Health Score (FHS)** is a **Prototype Observed Financial Health Indicator (FHI)** that computes a **0–100 score entirely from the user's logged transaction data** — no surveys, no self-report. It was not invented from scratch. Every design decision is grounded in an existing academic or institutional framework.

The FHS adapts the approach of the **CBA–Melbourne Institute Observed Financial Health Indicator** (Comerton-Forde et al., 2018), which uses actual bank-record data to derive a 0–100 observed scale with equal-weight components. We adapted the same transaction-based philosophy to a mobile SQLite database instead of bank records.

The Formula

FHS = Savings Rate + Overspend Control + Budget Adherence + Logging Consistency
(maximum 100 points — 25 points per component)

Component	Formula	Academic Basis
Savings Rate (25 pts)	25 × min(1, savingsRate / 0.20)	Warren & Tyagi (2005) — 50/30/20 rule; 20% savings as the standard personal finance target
Overspend Control (25 pts)	25 × (1 − weightedOverDays / activeDays)	Kahneman & Tversky (1979) — Prospect Theory: losses are more motivationally salient than equivalent gains
Budget Adherence (25 pts)	25 × (onBudgetCategories / totalBudgetCategories)	Financial Health Network FinHealth Score® Plan pillar; BSP CFIS (2025) — fewer than 50% of Filipino households maintain any written budget
Logging Consistency (25 pts)	25 × (loggedDays / activeDays)	Deci & Ryan (2000) — Self-Determination Theory: habit formation requires consistent, intrinsically motivated behavior

Adjustments (applied after the 4 components): - **Warning Decay:** −5 pts/day, max −15 pts — if user continues spending on a budget category after a warning. Grounded in Prospect Theory's loss-salience principle (Kahneman & Tversky, 1979). Applies to discretionary categories only; Bills, Health, Education are exempt. - **Gap Adjustment:** −3 pts/day unlogged (max −15); +2 pts/day confirmed no-spend (max +10). Grounded in behavioral consistency principle (Ariely, 2008; Duhigg, 2012).

Why Equal Weights (25 pts each)?

Three justifications:

1. **CBA–Melbourne Institute precedent** — their Observed FHI uses equal-weighted transaction-derived components summing to 100 (Comerton-Forde et al., 2018)
2. **OECD/JRC Handbook on Composite Indicators** (2008) — formally establishes equal weighting as the methodologically correct default when there is insufficient empirical basis for differential weights
3. **Transparency** — users can understand "each pillar is worth 25 points" without a finance degree

We do not have the multi-year longitudinal data needed to justify differential weights for this population. Equal weighting is the honest, defensible choice.

Alignment with Established Financial Health Frameworks

SmartSpend FHS Component	CBA–Melbourne Institute (2018)	FinHealth Score® (FHN)	CFPB Financial Well-Being Scale (2017)
Savings Rate	Observed savings behavior from bank records	Save pillar: savings rate and buffer	Q7: "I could handle a major unexpected expense"
Overspend Control	Observed overspending days (weighted)	Spend pillar: spending less than income	Q2: "My finances control my life" (reverse)
Budget Adherence	Category-level spending vs. limits	Plan pillar: on track with financial goals	Q9: "My finances are on track"
Logging Consistency	Data completeness metric	(Not directly measured)	(Not directly measured)

Key distinction: The CFPB scale is self-reported (users rate their own feelings). The Melbourne Institute and FinHealth Score use transaction data — the same approach as SmartSpend.

Why Each Component Was Chosen — Linked to Needs Assessment

Each component directly addresses a confirmed user problem from the preliminary needs assessment:

Problem Found in Needs Assessment	FHS Component
Difficulty saving money	Savings Rate — rewards users who save ≥20% of income
Regularly exceeds planned budget	Overspend Control — penalizes habitual overspending days
Fewer than half maintain any budget	Budget Adherence — incentivizes setting and following budgets
Inconsistent or no expense tracking	Logging Consistency — full score only when logging every active day

---

References

Comerton-Forde, C., Ip, E., Ribar, D. C., Ross, J., Salamanca, N., & Tsiaplias, S. (2018). *Using survey and banking data to measure financial wellbeing* (CBA–MI Technical Report No. 1). Commonwealth Bank of

Australia & Melbourne Institute.

Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits. *Psychological Inquiry, 11*(4), 227–268.

Financial Health Network. (2021). *FinHealth Score® Toolkit.* https://finhealthnetwork.org/tools/financial-health-score/

Gubbins, P., Mazzotta, B., & Rhyne, E. (2021). *Measuring financial health: Concepts and considerations.* UNSGSA Financial Health Working Group.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica, 47*(2), 263–292.

OECD & Joint Research Centre. (2008). *Handbook on constructing composite indicators.* OECD Publishing. https://doi.org/10.1787/9789264043466-en

Warren, E., & Tyagi, A. W. (2005). *All your worth: The ultimate lifetime money plan.* Free Press.

Bangko Sentral ng Pilipinas. (2025). *Consumer Finance Inclusion Survey (CFIS) 2025.* https://www.bsp.gov.ph/
